

```
#include <stdio.h>

int main(void)
{
    char a = 4;

    char b = -2;

    char e;

    printf("%d %d", ~++a + b, ~b ^ b << ~b);

    e = ~b && a % -b || b/a;

    printf(" %d",e);

    return 0;
}
```

☐ -8 -3 1

☐ -8 0 0

☐ Compile time error

☐ -8 -2 1

Which of the memory layout section given in options is also called as the 'bss' segment?

☐ Stack

☐ Uninitialized data segment

☐ Code segment

☐ Initialized data segment

What will be the output of the program given below?

```
#include <stdio.h>
int main()
{
    unsigned int k, m, a = 2390;
    unsigned int *ip = &a;
    char *p = (char *)ip;
    k = *p++;
    k = (~(~0 << 4) &*p) << 4 | ((15 << 4) &k) >> 4;
    m = *p << 3;
    printf("%d, %d", k, m);
    return 0;
}
```

☐ 133, 72

☐ 101, 72

☐ 101, 19120

☐ 149, 72

In which of the below given process(es) execution of one process depends on the other?

I) Independent process

II) Cooperative process

☐ Neither (I) nor (II)

☐ Both (I) and (II)

☒ Only (I)

☐ Only (II)

Which of the below given statement(s) is/are correct about the disadvantages of array implementation of linked list?

- I) The number of nodes required is not possible to predict.
- II) The number of nodes declared need not be remained allocated through out the execution.

☐ Only (II)

☐ Neither (I) nor (II)

☐ Both (I) and (II)

☐ Only (I)

What will happen if a system does not employ either a deadlock-prevention or a deadlock-avoidance algorithm?

- ☐ Nothing will happen.
- ☐ Deadlock situation may not arise.
- ☐ Deadlock situation may arise.
- ☐ Deadlock situation arise.

Which of the below given statement(s) is/are true

- I) Overridden methods can change return types; overloaded methods cannot, except in the case of covariance
- II) Object reference return types can accept null as a return value

☐ Only (II)

☐ Only (I)

☐ Both (I) and (II)

☐ Neither (I) nor (II)

```
#include<stdio.h>

int main()
{
    int x = 20, y = 5, a = 0x07, b = 24;

    if(x >>= y && x != 10)
        printf("%d", (a != 7) & x++ * y >> y);
    else
        printf("%d", a += 5 * a >> 3 / b++ | x * 4 * 2 - 3);

    return 0;
}
```

A clock interrupt handler on a system needs 2 msec/clock tick, wherein the clock runs at 60 Hz. What percentage of the CPU time is reserved for it?

☐ 1.2%

☐ 18.5%

☐ 12%

☐ 7.5%

```
#include <stdio.h>
```

```
void fun(char *p)
```

```
{
```

```
    const char *str = "Hello0";
```

```
    while((*p++ = *str++));
```

```
}
```

```
int main()
```

```
{
```

```
    int arr[10] = {'2'};
```

```
    char *ptr = (char*)arr;
```

```
    fun(ptr);
```

```
    printf("%d", ptr[0] + ptr[5]);
```

```
    return 0;
```

```
}
```

☐ 120

☐ 50

☐ 00

QUESTION: 29

GROUP: Programming Basics

SECTION: C

Which one of the given options is the correct 2's complement

$(-55)_{10}$

☐ 01010101

☐ 11001001

☐ 11101101

☐ 10110101

26 #include <stdio.h>

int fun(char);

27 int main(void)

{

28 const char a = 1;

fun(a);

29 printf("%d ", a);

30 ☆ return 0;

}

31 int fun(char a)

{

32 a = 1;

33 printf("%d ", a >> -1);

printf("%d ", 1 >> -1);

34 }

35 ☐ Compile time error

36 ☐ 0 2 1

37 ☐ 2 2 1

38 ☐ 0 0 1

39

QUESTION: 36 GROUP: 123456789101112131415161718192021222324252627282930313233343536373839404142434445464748495051525354555657585960616263646566676869707172737475767778798081828384858687888990919293949596979899100
In which type of rotations given in options where every node moves one position to left from the current position?

☐ Left Right Rotation

☐ Right Left Rotation

☐ Left Rotation

☐ Right Rotation

QUESTION: 41 Which of the allocation technique given in options allocates the smallest hole among the freely available partition i

☐ Best Fit

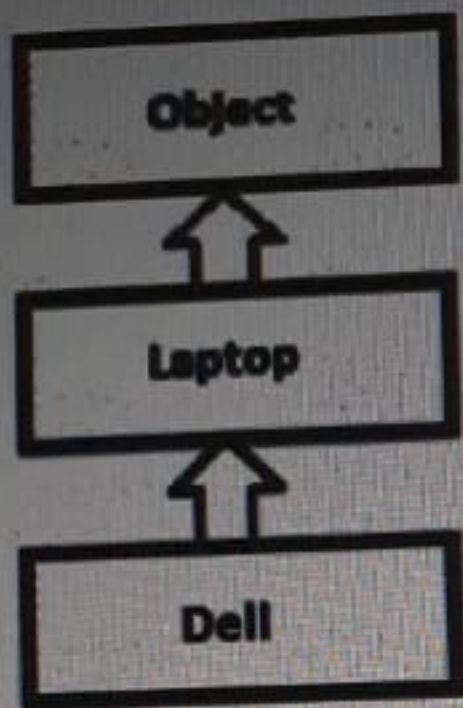
☐ Worst Fit

☐ Next Fit

☐ First Fit



Consider the image given below:



Which of the type of relationships in Inheritance is representing the above diagram?

☐ None of the given options

☐ USES - A

☐ IS-A

☐ HAS- A

What will be the output of the program given below?

```
#include <stdio.h>

int main()
{
    void *p;

    int NA[10];

    p = NA;

    printf("%d ", p);

    return 0;
}
```

- ☐ Compile time error
- ☐ The program will print the size of array 'NA'
- ☐ Run time time error
- ☐ The program will print the address of pointer 'p' in

What will be the output of the program given below?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int *j;
```

```
    void fun(int **);
```

```
    fun(&j);
```

```
    return 0;
```

```
}
```

```
void fun(int **k)
```

```
{
```

```
    int a = 100 * 1000 + 100 / 5;
```

```
    printf("%d", *k = a);
```

```
}
```

☐ 102100

☐ 200200

☐ 100005

An array $M[89][12]$ is initialized and stored as row-major order in memory in the form of a single dimensional array, Arr. What will be the index of $M[4][5]$ in Arr?

Note: Arr indexing starts from 0.]

☐ 53

☐ 48

☐ 20

☐ 64

QUESTION: 55

GROUP: Computer Science

SECTION: Data

The prefix expression given below is evaluated using stack.

$+ \ 2 \ * \ - \ * \ 5 \ 2 \ * \ / \ 12 \ \% \ 4 \ 3 \ 6 \ 1$

What will be the TOP of the stack after evaluation of '-'?

☐ 10

☐ 72

☐ -62

☐ 12

9 `#include <stdio.h>`

0 `int main(void)`

1 `{`

2 `char c = -1;`

3 `signed char d = 0;`

4 `printf("%d ", ~d);`

5 `printf("%d ", --d);`

6 `printf("%d ", d || c);`

7 `return 0;`

8 `}`

9 ☐ -1 -1 0

10 ☐ 1 -1 0

11 ☐ -1 -1 1

12 ☐ 1 1 1

What will be the output of the program given below?

```
#include <stdio.h>
int main()
{
    int max = 123, min = 10, *maxptr = &max, *minptr = &min;
    int **nptr = &minptr, **mptr = &maxptr;
    *maxptr = ++**mptr % **nptr;
    max -= (*minptr - **nptr && *maxptr || *minptr);
    printf("%d %d", ++**mptr, *minptr);
    return 0;
}
```

☐ 4 10

☐ 3 10

☐ 3 11

☐ 4 11

30

☐ Memory leak

31

☐ No logical error

32

☐ Invalid pointer usage

33

☒ Buffer overflow

34

23
int main() {

24 int arr[5];

25 for (int i = 0; i <= 5; i++) {

arr[i] = i * i;

26 }

printf("Array elements: ");

27 for (int i = 0; i < 5; i++) {

printf("%d ", arr[i]);

28 }

return 0;

29 }

30